

Sahil Katiyar

MUMBAI, MAHARASHTRA

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PROFESSIONAL SUMMARY

B.Sc. Data Science graduate building and evaluating production-grade AI systems - multi-agent RAG pipelines, MLOps on AWS, and empirical benchmarking of LLM agent techniques (RAGAS, LLM-as-judge). Comfortable across the stack: designing **agent architectures (LangGraph)**, **deploying them (Docker, Kubernetes, CI/CD)**, and proving they work with data, not assumptions. Seeking an AI Engineering internship to bring that build-then-measure approach to real-world products.

EDUCATION

Bachelors of Data Science and Analytics

Mumbai University (R.A. Podar College of commerce)

CGPA: 10.00 / 10.00

Graduation Year: 2026

SKILLS

Languages: Python, SQL

ML/DL: PyTorch, TensorFlow, Scikit-learn, Pandas, NumPy, NLP, Machine learning, Deep learning.

LLMs & Agents: LangChain, LangGraph, LangSmith, RAG, Prompt Engineering, Tool-Calling, Multi-Agent Orchestration, MCP, Context Engineering, Hugging Face, Transformers, Unsloth, LoRA/QLoRA Fine-tuning, Embeddings.

DevOps & Cloud: Docker, Kubernetes, AWS (EC2, S3, EKS, IAM), FastAPI, Git, GitHub Actions, CI/CD, MLflow, MLOps

Data & Retrieval: Vector Databases (Chroma), Neo4j, | **Evaluation:** RAGAS, LLM-as-a-Judge.

PROJECTS

Boss-Employee Agentic RAG System | [Github](#)

- Built a multi-agent AI system simulating an office workflow, where a **boss agent** dynamically assigns tasks to **employee agents (Data Analysis, Research)** powered by Groq, Gemini api.
- Designed an **Agentic Retrieval-Augmented Generation (RAG)** pipeline that adaptively routes queries across **SQL, Chroma, and Neo4j** based on the nature of the question, using vector search with reranking to improve retrieval accuracy on the Chromabased retrieval path. Enabled multi-step reasoning and planning using agent-based workflows to improve task execution and contextual response quality.
- Integrated **observability using LangSmith** to monitor errors, latency, and workflow execution, achieving ~5–15 seconds response time depending on query complexity.

CFPB Complaint Intelligence System — End-to-end MLOps pipeline | [Github](#)

- Built and deployed dual NLP models (PyTorch MLP, Scikit-learn MLP) for complaint product classification (**86% F1**) and urgency detection (**85% F1, 99% precision**).
- Designed an NLP preprocessing and feature engineering pipeline with **MLflow-based experiment tracking and model versioning. Automated containerized deployment to AWS EKS using Docker, Kubernetes, and a full CI/CD pipeline via GitHub Actions.** Troubleshoot and resolved production infrastructure issues (node sizing, memory limits, AWS account constraints) to ensure reliable model serving.

Context Engineering Agent – Benchmarking Retrieval Compression & Memory for LLM Agents | [Github](#)

- Built a LangGraph agent to test whether **context engineering** (filter -- dedupe -- re-rank via Cohere -- compress) improves performance, with **short-term (checkpointing)** and **long-term memory (Mem0)**.
- Benchmarked 7 pipeline variants using **LLM-as-judge and RAGAS metrics, cutting token usage ~44%** while holding answer quality nearly constant (4.39 to 4.28) vs. baseline. Validated the relevance filter under a real failure case, correctly dropping irrelevant search results before they reached the prompt.

AI-Driven Demand Forecasting & Supply Chain Optimization for Power Grid Infrastructure | [Github](#)

- Developed an AI-driven supply chain platform for power grid infrastructure, combining a **multi-output ML model with a multiagent GenAI system to automate procurement planning, logistics, and risk analysis.**
- Built an ExtraTreeRegressor model predicting multiple raw materials (steel, conductors, concrete) simultaneously, and designed a LangGraph-orchestrated multi-agent pipeline -- Price (Alpha Vantage, ExchangeRate-API), Logistics (Geoapify), and Risk (Google Gemini, SerpApi) agents.
- Integrated a conversational AI orchestrator powered by Llama-3-70B (via Groq) for real-time plan Q&A, and deployed via Docker for production-ready execution, achieving ~5–20 seconds response time.

CO-CURRICULAR/CERTIFICATES

- Google's Genai Exchange Hackathon:** worked with **Google Cloud** and made my first gen AI project - RAG.
- Led CodeCrux, the college's first inter-university coding competition, managing the organizing team.